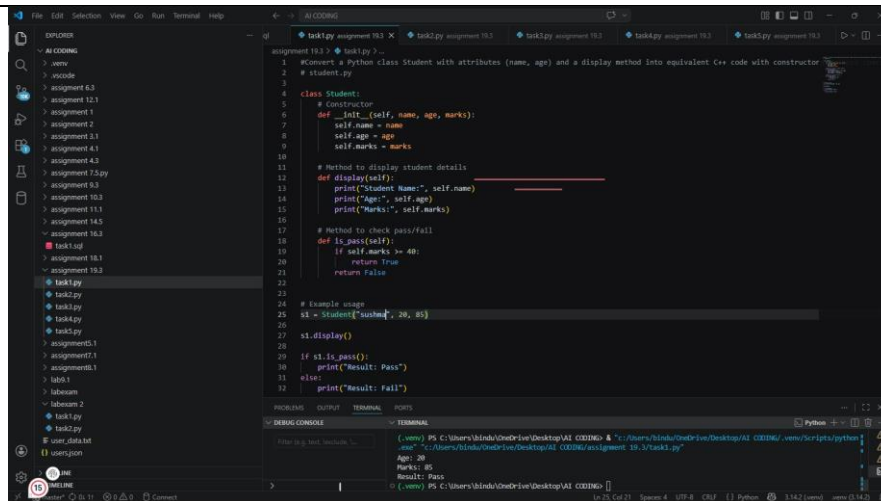


SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
Instructor(s) Name			
CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
NAME	M.Anu Chandra	HALLTICKET:	2303A54070
Duration	2 Hours	Batches	48
Assignment Number:19.3(Present assignment number)/24(Total number of assignments)			
Q.No.	Question	Expected Time to complete	
1	Lab 19 – Code Translation: Converting Between Programming Languages Lab Objectives: - Understand how AI tools can assist in translating code between different programming languages. - Learn to verify correctness and functionality after translation. - Explore syntactic and semantic differences between languages (e.g., Python, Java, C++). - Practice debugging and optimizing AI-translated code.	Week10 - Wednesday	
	Task 1 – Python to C++ Conversion Task: Translate a simple Python class into C++ using AI assistance. Scenario A college system stores student details (name, age) and displays them. Convert this logic from Python to C++. Prompt Used Convert a Python class Student with attributes name and age into equivalent C++ code with constructor and display method.		



```
1 # Convert a Python class Student with attributes (name, age) and a display method into equivalent C++ code with constructor
2 # student.py
3
4 class Student:
5     # Constructor
6     def __init__(self, name, age, marks):
7         self.name = name
8         self.age = age
9         self.marks = marks
10
11     # Method to display student details
12     def display(self):
13         print("Student Name:", self.name)
14         print("Age:", self.age)
15         print("Marks:", self.marks)
16
17     # Method to check pass/fail
18     def is_pass(self):
19         if self.marks >= 40:
20             return True
21         return False
22
23
24 # Example usage
25 st = Student("sushant", 20, 85)
26
27 st.display()
28
29 if st.is_pass():
30     print("Result: Pass")
31 else:
32     print("Result: Fail")
33
34
```

Output:

```
Student Name: sushant
Age: 20
Marks: 85
Result: Pass
```

Explanation

This program creates a student class and stores name and age. The display function prints the values. In C++, we use constructor and access specifiers unlike Python.

Task 2 – Java to Python Function Conversion

Task:

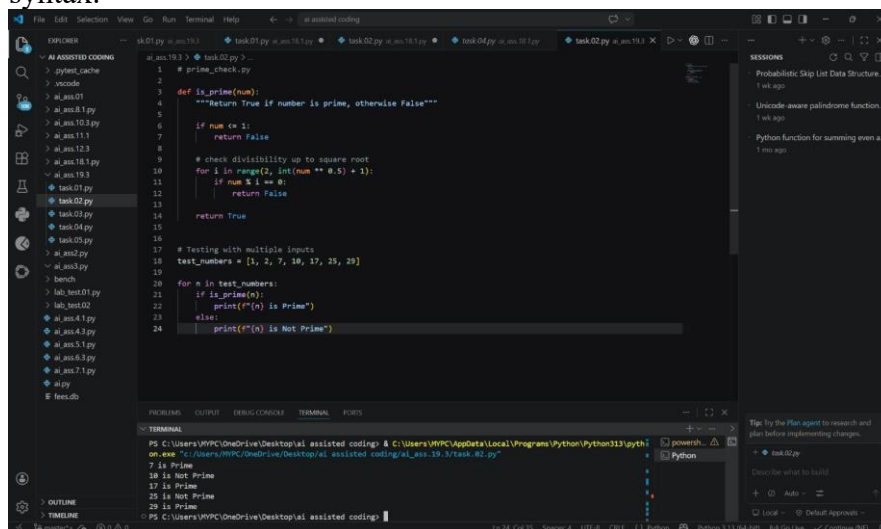
Convert a Java method that checks whether a number is prime into Python.

Scenario

A system checks whether a number is prime. Convert Java method into Python.

Prompt Used

Convert Java method isPrime into Python function using simple syntax.



```
1 # prime_check.py
2
3 def is_prime(num):
4     """Return True if number is prime, otherwise False"""
5
6     if num <= 1:
7         return False
8
9     # check divisibility up to square root
10    for i in range(2, int(num ** 0.5) + 1):
11        if num % i == 0:
12            return False
13
14    return True
15
16
17 # Testing with multiple inputs
18 test_numbers = [1, 2, 7, 10, 17, 25, 29]
19
20 for n in test_numbers:
21     if is_prime(n):
22         print(f"{n} is Prime")
23     else:
24         print(f"{n} is Not Prime")
25
26
```

Output:

```
2 is Prime
7 is Prime
10 is Not Prime
17 is Prime
25 is Not Prime
29 is Prime
```

Explanation

The program checks if a number is divisible by any number other than 1 and itself. If yes, it is not prime.

Task 3 – Pseudocode to Python Implementation

Task:

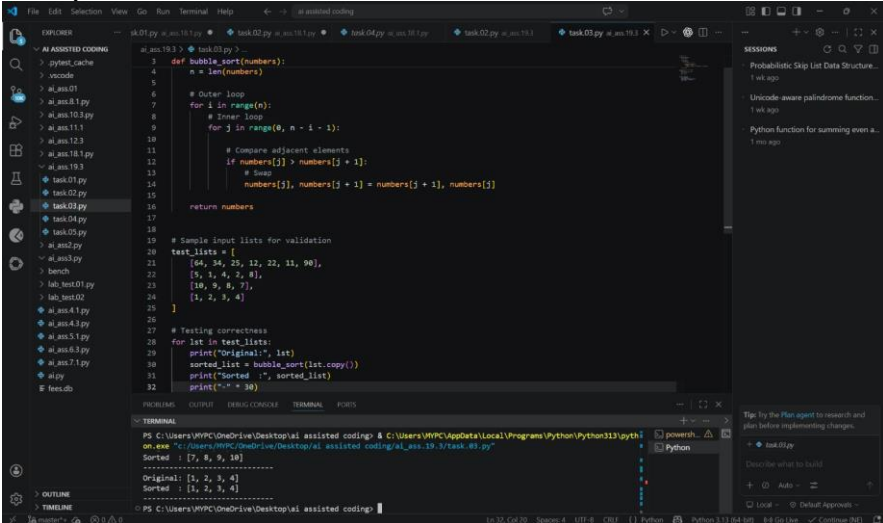
Translate a given pseudocode algorithm into Python using AI.

Scenario

Sort a list of numbers using bubble sort logic.

Prompt Used

Convert bubble sort pseudocode into Python program with example.



Explanation

Bubble sort compares adjacent elements and swaps them if they are in wrong order until the list is sorted.

Task 4 – SQL to Pandas Query

Task:

Translate a **SQL query** into a **Pandas DataFrame operation** in Python.

Scenario

Convert SQL query into Python Pandas code.

Prompt Used

Convert SQL SELECT query with condition into Pandas filtering.

The screenshot shows a VS Code editor with a Python file named `task04.py`. The script creates a sample employee dataset and filters it based on salary. The terminal output shows the original dataset and the filtered result.

```
1 import pandas as pd
2
3 # Creating sample employee data
4 data = {
5     "name": ["Uday", "Rahul", "Anita", "Sneha", "Kiran"],
6     "age": [22, 30, 27, 25, 35],
7     "salary": [45000, 60000, 52000, 48000, 75000]
8 }
9
10 employees = pd.DataFrame(data)
11
12 print("Original DataFrame:")
13 print(employees)
```

Terminal Output:

```
Original DataFrame:
   name  age  salary
0  Uday  22  45000
1  Rahul  30  60000
2  Anita  27  52000
3  Sneha  25  48000
4  Kiran  35  75000
```

Explanation

The program filters rows where salary is greater than 50000 using Pandas similar to SQL WHERE clause.

Task 5 – Real-Time Application: Algorithm Translation Across Languages

Scenario:

A company maintains algorithms written in different programming languages.

Prompt Used

Translate a Python linear search algorithm into equivalent C code. Ensure both versions are executable and produce the same output. Highlight syntax, memory management, and library differences.

The screenshot shows a VS Code editor with a C file named `task05.py`. The program implements a linear search algorithm in C. The terminal output shows the execution of the program with a test case.

```
1 # linear_search.py
2
3 def linear_search(arr, key):
4     for i in range(len(arr)):
5         if arr[i] == key:
6             return i # return index
7     return -1
8
9 # Test execution
10 numbers = [10, 25, 30, 40, 50]
11 key = 30
12
13 result = linear_search(numbers, key)
14
15 if result != -1:
16     print("Element found at index:", result)
17 else:
18     print("Element not found")
```

Terminal Output:

```
4 Kiran 35 75000
PS C:\Users\WPC\OneDrive\Desktop\ai assisted coding> C:\Users\WPC\AppData\Local\Programs\Python\Python313\python.exe C:\Users\WPC\OneDrive\Desktop\ai assisted coding\ai_ass_19_3\task-05.py
Element found at index: 2
PS C:\Users\WPC\OneDrive\Desktop\ai assisted coding> C:\Users\WPC\AppData\Local\Programs\Python\Python313\python.exe C:\Users\WPC\OneDrive\Desktop\ai assisted coding\ai_ass_19_3\task-05.py
Element found at index: 2
PS C:\Users\WPC\OneDrive\Desktop\ai assisted coding>
```

	Short Explanation : This program searches for an element in a list using linear search. It checks each element one by one until it finds the key. The same logic is implemented in both Python and C, but the syntax and memory handling differ.	
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